Encoder-Decoder Contrast for Unsupervised Anomaly Detection in Medical Images

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***Abstract*— Unsupervised anomaly detection (UAD) aims to recognize anomalous images based on the training set that contains only normal images. In medical image analysis, UAD benefits from leveraging the easily obtained normal (healthy) images, avoiding the costly collecting and labeling of anomalous (unhealthy) images. Most advanced UAD methods rely on frozen encoder networks pre-trained using ImageNet for extracting feature representations. However, the features extracted from the frozen encoders that are borrowed from natural image domains coincide little with the features required in the target medical image domain. Moreover, optimizing encoders usually causes pattern collapse in UAD. In this paper, we propose a novel UAD method, namely Encoder-Decoder Contrast (EDC), which optimizes the entire network to reduce biases towards pre-trained image domain and orient the network in the target medical domain. We start from feature reconstruction approach that detects anomalies from reconstruction errors. Essentially, a contrastive learning paradigm is introduced to tackle the problem of pattern collapsing while optimizing the encoder and the reconstruction decoder simultaneously. In addition, to prevent instability and further improve performances, we propose to bring globality into the contrastive objective function. Extensive experiments are conducted across four medical image modalities including optical coherence tomography, color fundus image, brain MRI, and skin lesion image, where our method outperforms all current state-of-the-art UAD methods. Code is available at: https://github.com/guojiajeremy/EDC**

***Index Terms*— Medical anomaly detection, unsupervised learning, feature reconstruction, contrastive learning, anomaly localization.**

I. Introduction

DEEP neural networks have become a popular methodology for analyzing medical images [1], [2], [3]. These successes should be attributed not only to the profound progress in deep learning techniques but also to a great number of carefully annotated medical image datasets [4]. However, the collection and labeling of medical images is usually an arduous task, especially for rare diseases [5]. It is also difficult to exhaust every possible anomaly for building disease screening systems. In health examinations, the amount of normal (healthy) subjects usually overwhelms anomalous (unhealthy) subjects, making normal image collection much easier. Unsupervised anomaly detection (UAD) aims to identify anomalous images based on the training set that contains only normal images. Recently, UAD has attracted increasing interest in medical image analysis [5], [6], [7].

Most UAD methods attempt to learn the distribution of available normal samples. Pixel reconstruction methods [5], [7], [8], [9], [10] train auto-encoders or generative adversarial networks (GAN) [7] from scratch to reconstruct normal images, relying on the hypothesis that the networks trained on normal images can only reconstruct normal regions, but fail for anomalous regions. However, recent studies [11], [12] suggest that anomalous regions can also be well constructed when normal and anomalous regions share similar pixel values or anomalies are too subtle. Accordingly, feature reconstruction methods reconstruct features from pre-trained encoders instead of raw pixels, because learned features provide more discriminative and noise-free representation [11], [12], [13]. To prevent the reconstruction networks from converging to a trivial solution, i.e., pattern collapse, the parameters of the encoders should be frozen during training [11]. However, the semantic gap between natural images (usually ImageNet [14]) on which the frozen networks were pre-trained and medical images in various modalities causes poor transfer ability. Memory matching methods [15], [16], [17], [18] memorize all anomaly-free features extracted from pre-trained networks in training, and match them with test samples during inference. Apart from the expensive computational overhead, these methods also struggle in transfer as the feature extractor is not optimized in the target domain.

In this study, our motivation is to reorient the large-scale pre-trained encoders to target medical UAD datasets without losing their ability to extract discriminative features. We propose a novel UAD method, namely Encoder-Decoder Contrast (EDC), as illustrated in Fig. 1(c). The essence of contrastive learning (Fig. 1(a)) and feature reconstruction UAD (Fig. 1(b)) are elegantly combined, orienting the network in the medical domain by training all parameters end-to-end.

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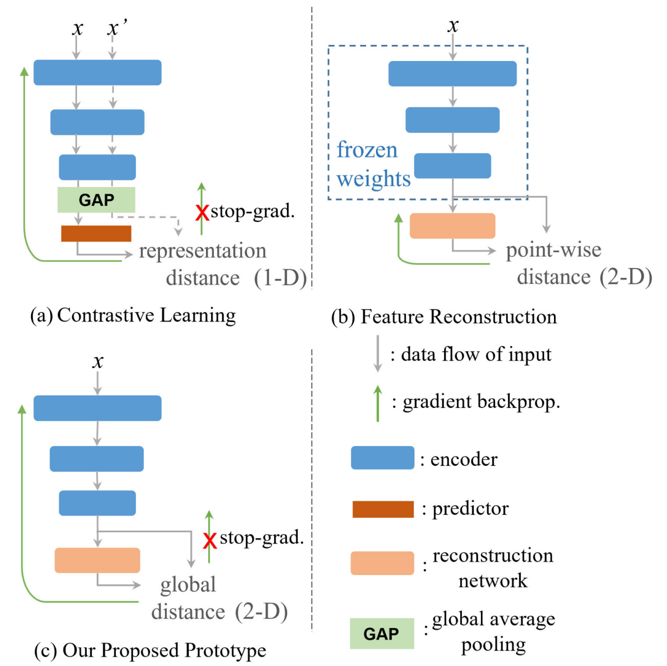


Fig. 1. Comparison of three similar architectures. (a) Contrastive learning [19], [20] for self-supervised representation learning. It maximizes the similarity between the representations of two augmentations of one image (x and x') while trying to avoid collapsing solutions (all inputs encoded to a constant). (b) Feature reconstruction [11], [13]. It reconstructs 2-D features extracted by frozen pre-trained encoder for anomaly detection. (c) Our proposed EDC prototype. It combines 2-D reconstruction with the elements of contrastive learning, i.e. training encoder, stop-gradient, and global distance. Here, 1-D and 2-D indicate the distance is calculated by 1-D feature representations R^C and 2-D feature map $R^{C\times H\times W}$, respectively.

First, we start with the feature reconstruction-based UAD methods that detect anomalies by reconstruction errors between the encoder and the reconstruction network (decoder) [11], [13]. Second, we explain why the direct optimization of encoder causes feature collapse by probing the behavior of encoder features. Accordingly, we introduce the key element of contrastive learning into feature reconstruction, building a feature map contrast paradigm to train the encoder and decoder jointly on medical images. Specifically, we target the essential operation, stop-gradient, that enables contrastive representation learning methods [19], [20] to train without collapse. Third, we observe that optimizing the encoder makes the training procedure extremely unstable, which causes degradation in the end. Inspired by global average pooling (GAP), a new optimization objective, namely global cosine distance, is proposed to make the contrast in a global manner to stabilize the optimization and further improve performances.

We evaluate EDC with extensive experiments on four medical image datasets, including optical coherence tomography (OCT) [21], color fundus image [22], brain MRI [23] and skin lesion image [24]. Without elaborately designed modules or pseudo anomalies, our proposed method achieved superior performance compared with previous state-of-the-arts. On the widely used OCT2017 benchmark, proposed method achieves nearly perfect scores (AUC=99.56%), reducing the error of previous best by three-quarters. The contributions of this study can be summarized as:

1) To address the poor transfer performance of the frozen pre-trained encoder, we propose a simple yet effective UAD method for detecting medical anomalies, combining the key elements of contrastive learning and feature reconstruction.
2) We introduce the stop-gradient operation to optimize the entire network in target medical domains without feature collapse.
3) We design a new objective loss function, introducing globality into cosine distance to prevent instability and divergence.

II. Related Work

A. Unsupervised Anomaly Detection

Unsupervised anomaly detection methods rely on learning the distribution of available normal samples. Classical approaches focus on utilizing machine learning methods for building a one-class classifier, such as one-class support vector machine (OC-SVM) [25] and support vector data description (SVDD) [26]. With the advance of deep learning, most recent works apply neural networks on UAD tasks. Reconstruction-based methods are under the hypothesis that the networks trained on normal images can only reconstruct anomaly-free regions, but fail for anomalous regions [27]. Within the reconstruction paradigm, pixel reconstruction methods are trained from scratch to reconstruct normal pixels via auto-encoder (AE) [10], [27], variational auto-encoder (VAE) [9], [28], and generative adversarial network (GAN) [5], [7], [29]. Such methods detect anomalies by the pixel value errors between input images and reconstructed images. Recent studies suggest that unseen anomalous regions can also be well restored when normal and anomalous regions share similar pixel values or the anomalies are too subtle [11], [13].

Most advanced methods utilize networks pre-trained on large-scale datasets (ImageNet [14]) for extracting discriminative and informative feature representation in UAD tasks. Because the available UAD training set is usually small and contains only one class of image, i.e., normal samples, the usage of a powerful pre-trained network alleviates the need for training networks from scratch with limited supervision. Feature reconstruction methods reconstruct features from pre-trained encoders instead of raw pixels [11], [12], [30]. To prevent the reconstruction networks from converging to a trivial solution, i.e., pattern collapse, the parameters of the encoders are frozen during training [13]. Feature distillation is a special configuration of feature reconstruction, where the student network mimics the output features of the pre-trained teacher network with the same input of normal images [11], [31], [32]. The above two approaches are also based on the hypothesis that the models trained on normal samples only succeed in predicting features of normal regions. Memory matching methods [15], [16], [17] memorize all (or their reduced subset) normal features extracted from pre-trained encoders and match them with test samples during inference. Since these methods require memorizing, processing, and matching nearly all features from training samples, they are computationally expensive in both training and inference, especially when the training set is large. Pseudo-anomaly methods generate artificial defects on normal images to imitate anomalous images, converting UAD to supervised classification [33] or segmentation task [34]. These methods deeply rely on how well the pseudo anomalies match real anomalies [11], which makes it hard to generalize to different datasets. Another type of paradigm, i.e., normalizing-flow, models the likelihood (distribution) function of normal samples. Such methods [35], [36], [37] deploy a sequence of invertible and differentiable mappings to transform a complex distribution of extracted image features into a standard normal distribution, and detect anomalies by estimating the likelihood of features.

There are considerable semantic gaps between large-scale natural images on which the frozen networks were pre-trained and the medical images in various modalities. Therefore, methods based on pre-trained networks struggle in transferring to medical images as the feature encoders are not optimized in the target medical domain. There are some efforts [6], [38], [39] paying attention to first pre-train a domain-specific feature extractor on normal samples via self-supervised learning and then run aforementioned UAD methods. However, such two-step approaches require an additional pre-text training phase and a carefully designed pre-text task, which limits efficiency and generality. In this work, we mainly focus on orienting the network in the medical domain during UAD training.

B. Anomaly Detection in Medical Images

UAD methods for medical image are mostly adaptations or combinations of previously discussed approaches. AnoGAN [8] and f-AnoGAN [7] utilize GAN to detect anomalies in medical images. VAE is adopted for UAD on ultra-sound images [28]. A hybrid framework SALAD [5] was proposed to combine GAN and pseudo-anomaly. Normal images are first augmented to carry the artificial anomaly through pixel corruption and pixel shuffling. The fake abnormal images and normal ones are fed to the GAN for learning discriminative feature representations. This method demands strong prior knowledge of the real anomaly type. Most recently, SQUID [40] was proposed for detecting anomalies in chest x-ray, which is a GAN-based in-painting network with memory bank. ProxyAno [41] makes use of an auto-encoder with a memory bank in the middle to reconstruct super-pixels instead of raw pixels. STMC [42] proposed to memorize texture and structure information to avoid reconstructing anomalies too well. SCVAE [43] is proposed for retinal OCT images by introducing both spatial and contextual constraints in the latent space of VAE to amplify the anomalous reconstruction errors. The above methods are trained from scratch, without the help of pre-trained feature extractors. Self-supervised learning [44] is used to pre-train networks on medical images for better transfer [6] as supplementary to the base UAD methods. AE-flow [37] was proposed as a normalizing flow-based AE for modeling the distribution of features extracted by an ImageNet encoder. The method also suffers from the transfer ability of the frozen pre-trained feature encoder.

III. Method: From Reconstruction to Contrast

First, we give the definition of unsupervised anomaly detection. During training, a dataset with N normal images is available, given as $\mathcal{D}^{train} = \left\{(x_i, y_i)\right\}_{i=1}^{N}$, $y_i \in \{0\}$, where x_i is the i^{th} sample with its label y_i. During test, $\mathcal{D}^{test} = \left\{(x_i, y_i)\right\}_{i=1}^{M}$, $y_i \in \{0, 1\}$ (0 for normal, 1 for anomalous) containing both normal and anomalous images is given. The goal is to build a model using $\mathcal{D}^{train}$ to distinguish $(x, 1)$ from $(x, 0)$ in $\mathcal{D}^{test}$, while localizing anomalous regions as explanations. As usual, no validation set is used in UAD settings because no anomalous samples are available in the training data which includes the validation set.

Instead of introducing our proposed method directly, we present an exploration journey starting from the conventional feature reconstruction (*config.* 1) to our Encoder-Decoder Contrast with global cosine distance (EDC, *config.* 4) in a *problem and solution* manner. We believe such exploration will motivate researchers to rethink the role of encoders in UAD. In this section, the exploration is conducted on the APTOS dataset. By convention, area under the receiver operating characteristic (AUC) is used as the primary evaluation metric.

A. Config. 1: Feature Reconstruction, A Baseline

First, we build a simple feature reconstruction-based UAD approach, encoder-decoder reconstruction, which is the simplified version of RD4AD [11] without the multi-scale feature bottleneck. This configuration consists of a pre-trained encoder and a reconstruction decoder, as depicted in Fig. 2(a). The encoder is frozen, aiming to extract informative feature maps with different spatial sizes. The decoder is the reverse of the encoder, i.e., the down-sampling operation at the beginning of each layer is replaced by up-sampling. During training, the decoder learns to reconstruct the output of the encoder layers by maximizing the similarity between feature maps. During inference, the decoder accurately reconstructs anomaly-free regions of the feature maps but fails for anomalous regions as the decoder has never seen such samples. Without loss of generality, we consider the standard 4-layer ResNet [45] as our encoder and reconstruct the output of the first three layers. Mathematically, let $f_E^k, f_D^k \in \mathbb{R}^{C^k \times H^k \times W^k}$ denote the output feature maps from the k-th layer of the encoder and decoder respectively, where C^k, H^k, and W^k denote the number of channels, height, and width of the k-th layer, respectively. As illustrated in Fig. 2(e), the point-wise cosine distance [11], [32] is used to measure the difference between f_E^k and f_D^k point by point, obtaining a 2-D anomaly map $\mathcal{M}^k \in \mathbb{R}^{H^k \times W^k}$:

$$\mathcal{M}^k(h, w) = 1 - \frac{f_E^k(h, w)^T \cdot f_D^k(h, w)}{\left\| f_E^k(h, w) \right\| \left\| f_D^k(h, w) \right\|}, \tag{1}$$

where $\|\cdot\|$ is ℓ_2 norm, and a set of (h, w) locates a spatial point in the feature map. A large value in $\mathcal{M}^k$ indicates anomaly at the corresponding position. Final anomaly map $\mathcal{M}^{ano}$ is obtained by up-sampling all $\mathcal{M}^k$ to the same size and makes an average, as:

$$\mathcal{M}^{ano} = \sum_{k=1}^{3} \mathcal{R}(\mathcal{M}^k), \tag{2}$$

where $\mathcal{R}$ denotes the up-sampling operation. The maximum or mean value of $\mathcal{M}^{ano}$ is taken as the anomaly score for the

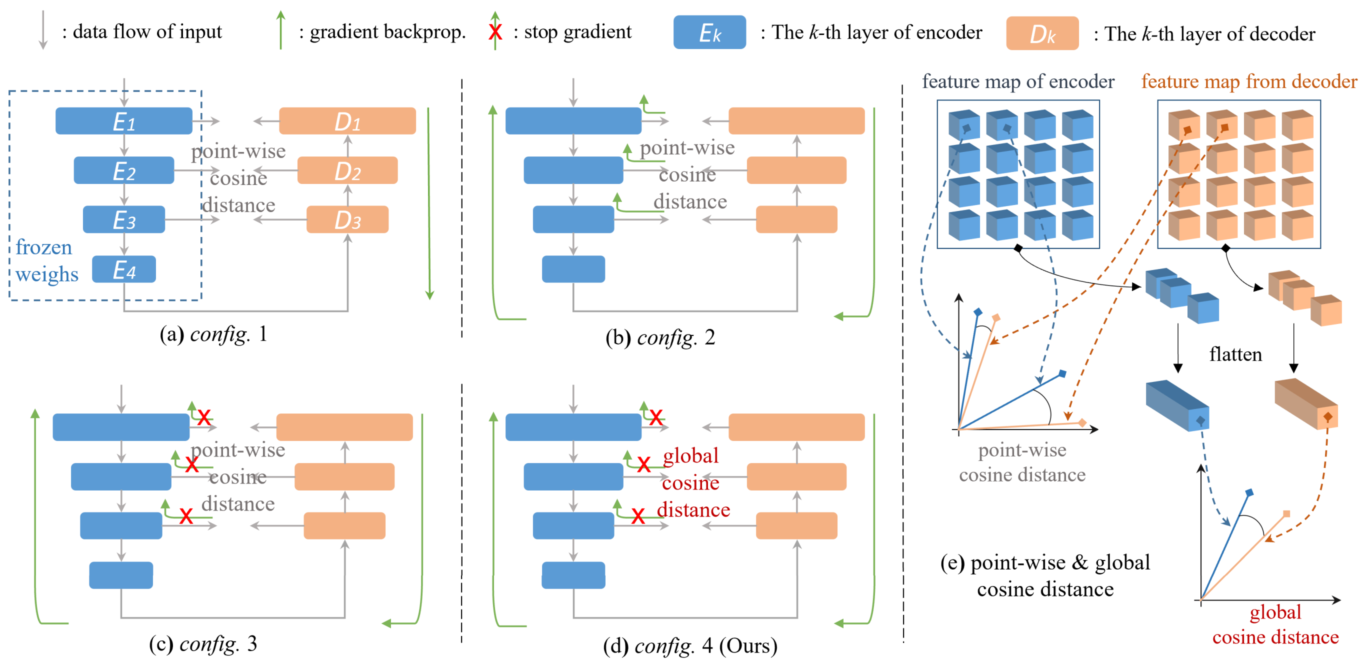


Fig. 2. Training configurations. (a) *config.* 1, encoder-decoder reconstruction. (b) *config.* 2, encoder-decoder reconstruction without freezing encoder. (c) *config.* 3, encoder-decoder contrast. (d) *config.* 4 (Ours), encoder-decoder contrast with global cosine distance. (e) Illustration of point-wise cosine distance and global cosine distance.

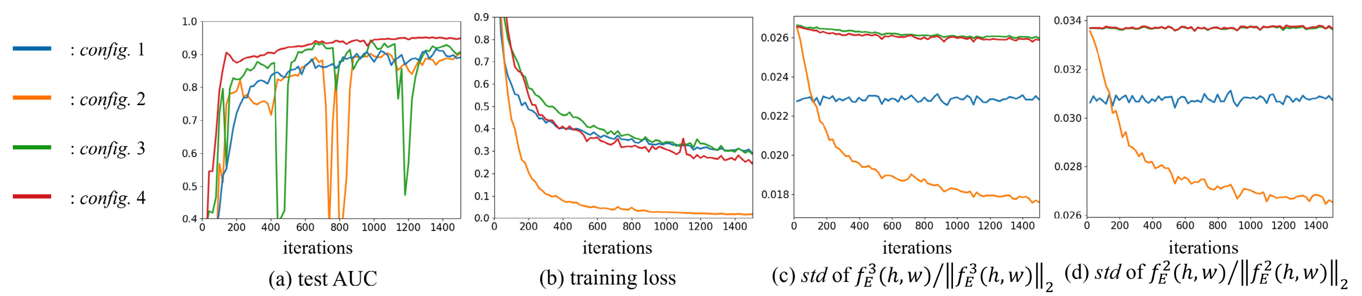


Fig. 3. Training record. (a) Test AUC. (b) Training loss. (c) Feature diversity of the 3rd encoder layer. (d) Feature diversity of the 2nd encoder layer.

image. During training, the anomaly maps are minimized on normal images by the point-wise cosine distance loss, given as:

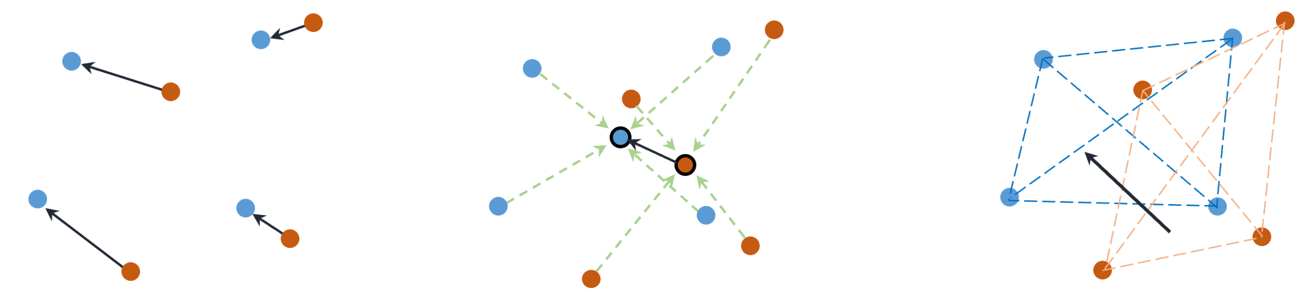
$$\mathcal{L} = \sum_{k=1}^{3} \frac{1}{H^k W^k} \sum_{h=1}^{H^k} \sum_{w=1}^{W^k} 1 - \frac{f_E^k(h,w)^T \cdot f_D^k(h,w)}{\left\| f_E^k(h,w) \right\| \left\| f_D^k(h,w) \right\|}. \tag{3}$$

The AUC of test set is presented by the blue line in Fig. 3(a), indicating that *config.* 1 can serve as a solid baseline.

B. Config. 2: Unfreeze Encoder

It seems straightforward to adapt the network in the target medical domain by jointly optimizing the encoder and decoder, as illustrated in Fig. 2(b). However, previous studies suggest that training encoders would lead encoders to extract indiscriminative features that are easy to reconstruct [13], which is also described as pattern collapsing or converging to trivial solution [11]. In this configuration, we observe a similar phenomenon that the training loss quickly decreases to nearly zero, as shown in Fig. 3(b). Though the network converges to an easy solution, the AUC does not degrade completely as previously expected but rises slowly to a comparable result as in Fig. 3(a).

To inspect the behavior of the encoder feature, we examine the diversity of feature maps by probing the standard deviation (*std*) of $f_E^k(h,w)/\left\|f_E^k(h,w)\right\|_2$. The larger the *std*, the more discriminative the feature, and the more distinctive between normal and anomalous regions. We plot the feature diversity of the 2nd and 3rd encoder layers during training in Fig. 3(c) and Fig. 3(d), respectively. The *std* of *config.* 2 drops quickly during training, while the *std* of *config.* 1 keeps relatively constant as a comparison. In addition, the *std* of *config.* 2 does not completely collapse to zero, which explains the passable AUC performance. The experiment shows that the trivial solution is attributed to the degeneration of the features extracted by the optimized unfrozen encoder. It is noted that the *std* starts from different values because of the different modes of batch normalization (BN) [46] during training. BN is



(a) point-wise cosine distance (b) cosine distance after GAP (c) global cosine distance

Fig. 4. Optimization of cosine distance losses. The black arrows represent optimization directions.

in eval mode in *config.* 1 using pre-trained running mean and variation, and it is in train mode in other configurations using batch mean and variation.

C. Config. 3: Encoder-Decoder Contrast

Contrastive learning [19], [20], [44] for self-supervised representation learning (SSL) maximizes the similarity between the representations of two augmentations of one image while trying to avoid collapsing solutions (all inputs are encoded to a constant value). It has been proved that the stop-gradient operation plays an essential role in preventing collapsing for SSL methods that rely only on positive pairs [19]. Interestingly, we found that contrastive methods can be explained from the perspective of feature reconstruction. Take SimSiam [19] as an example, the predictor (a multi-layer perceptron, MLP) can also be regarded as a reconstruction network that reconstructs the representation of one view x to match the representation of another view x'. Vice versa, feature reconstruction UAD can be converted to contrastive learning as well, regarding the reconstruction network as a predictor. Literally, "reconstruction" means the reconstructed object is constant, while the objects of "contrast" are adaptive during training.

To this end, we introduce the stop-gradient operation into encoder-decoder reconstruction, transforming it into a contrastive learning paradigm. In this configuration, the network is optimized by contrasting the feature maps of the encoder and decoder, as shown in Fig. 2(c). The gradients propagate, instead of directly into the encoder, back into the encoder through the decoder. It is implemented by modifying (3) as:

$$\mathcal{L}=\sum_{k=1}^{3}\frac{1}{H^kW^k}\sum_{h=1}^{H^k}\sum_{w=1}^{W^k}1-\frac{sg\left(f_E^k\left(h,w\right)^T\right)\cdot f_D^k\left(h,w\right)}{sg\left(\left\|f_E^k\left(h,w\right)\right\|\right)\ \left\|f_D^k\left(h,w\right)\right\|},\tag{4}$$

where sg denotes the stop-gradient operation. The encoder features are treated as constants in (4), but still optimized by the gradients of $\mathcal{L}$ from the decoder. The training of this configuration can be seen as a mutual reinforcement. The optimization of f_D^k makes both the encoder and decoder more informative in the task image domain, while the more informative f_E^k requires further optimization of f_D^k. In Fig. 3(c) and (d), we plot the feature *std*, showing that *config.* 3 can constantly extract more discriminative features. The performance rises quickly at the beginning of the training; however, we observe extreme instability during the training with "dips" in the AUC curve, as shown by the green curve in Fig. 3(a). The high and stable feature diversity indicates that this instability is not caused by the degradation of features as in *config.* 2.

D. Config. 4: Global Cosine Distance

We are now one step away from the last success. The *config.* 3 proves to be effective to extract discriminative feature representation on the target domain by contrasting. However, the instability in training limited its performance, especially in the UAD setting where validation set is not available for selecting the best checkpoint. In addition, previous studies suggest that the instability in neural network training harms the final performance, though catastrophic failure (e.g., divergence) is not caused [47]. Therefore, it is of great importance to find the reason and the solution to the instability.

Recalling the structure of SSL methods, there is still one component missing: globality. It is a convention to use global average pooling (GAP) to pool the 2-D feature map of the last network layer to a 1-D representation, in both supervised classification and SSL approaches. In UAD, the use of GAP will simply mess up all feature points together, losing the ability to distinguish normal and anomalous regions, as shown in Fig. 4(b). Therefore, we try to bring globality into the loss function directly. Let $\mathcal{F}$ denote a flatten operation that casts a 2-D feature map $f\in\mathbb{R}^{C\times H\times W}$ to a vector $\overline{f}\in\mathbb{R}^{CHW}$. Our proposed global cosine distance loss is given as:

$$\mathcal{L}_{global}=\sum_{k=1}^{3}1-\frac{sg\left(\mathcal{F}\left(f_E^k\right)^T\right)\cdot\mathcal{F}\left(f_D^k\right)}{sg\left(\left\|\mathcal{F}\left(f_E^k\right)\right\|\right)\ \left\|\mathcal{F}\left(f_D^k\right)\right\|}\tag{5}$$

The calculation difference between $\mathcal{L}_{global}$ and point-wise $\mathcal{L}$ is depicted in Fig. 2(e). In $\mathcal{L}_{global}$, a variation of one feature point will alter the whole representation vector and further affects the cosine distance, as opposed to the point-by-point independent cosine distance in $\mathcal{L}$. Intuitively, $\mathcal{M}^k$ (1) is also minimized along with the minimization of $\mathcal{L}_{global}$. It is easy to prove that minimizing global cosine distance to 0 equals to minimizing $\mathcal{M}^k$ to 0. For simplicity, we denote $\mathcal{F}\left(f_E^k\right)$ by $A=[a_1,\ldots,a_N]$, and $\mathcal{F}\left(f_D^k\right)$ by $B=[b_1,\ldots,b_N]$, where each a_i and b_i represent feature points $f_E^k\left(h,w\right)$ and $f_D^k\left(h,w\right)$

at the same location. Therefore, the $\mathcal{L}_{global}$ of the k-th layer can be given as:

$$1 - \frac{A^T \cdot B}{\|A\| \ \|B\|} = \frac{1}{2}\left(\frac{A}{\|A\|} - \frac{B}{\|B\|}\right)^2 = \frac{1}{2}\left\|\frac{A}{\|A\|} - \frac{B}{\|B\|}\right\|.$$

If the above equation equals to 0,

$$\frac{A}{\|A\|} = \frac{B}{\|B\|}, \quad \frac{a_i}{\|A\|} = \frac{b_i}{\|B\|}, \ \forall i \in 1 \ldots N.$$

As a_i and b_i are linearly correlated, their cosine distance is 0. Thus, $\mathcal{M}^k$ is 0 on each location. Though losses are not completely optimized to 0 in neural network training, we still use point-wise $\mathcal{M}^{ano}$ as anomaly map just like previous configurations.

The AUC record is shown in red in Fig. 3(a), exceeding all previous configurations with stability. We coarsely make a hypothesis on how $\mathcal{L}_{global}$ work: the loss landscape of $\mathcal{L}_{global}$ is flatter than $\mathcal{L}$, especially when both the encoder and decoder are included. In *config.* 2 and 3, the "dips" in the AUC curve are sometimes accompanied by "spikes" in the training loss curve, as shown in Fig. 3, which implies the optimizer encounters a cliff in the loss landscape. The rough loss landscape of $\mathcal{L}$ can be caused by the point-by-point distance, as depicted in Fig. 4(a). The fitting of one region may cause under-fitting of the others, and further disturbs the optimization. $\mathcal{L}_{global}$ can be seen as the distance between manifolds of feature points, as depicted in Fig. 4(c), which measures the consistency globally instead of excessively focusing on individual regions. The real mechanism can be more complicated and is still under further investigation.

IV. Experiments

A. Datasets

OCT2017 is an retinal optical coherence tomography (OCT) dataset [21]. The dataset is labeled into four classes: normal, drusen, DME, and CNV. The latter three classes are considered anomalous. The dataset was already separated into training and test set. The official training set contains 26,315 normal images and the official test set contains 1,000 images. It is noted that the 2-D images are slices from the original 3-D imaging modality. Images are resized to 256 × 256 pixels.

APTOS is a color fundus image dataset, available as the official training set of 2019 APTOS blindness detection challenge [22]. The dataset contains 3662 images with annotated grade 0-4 to indicate different severity of diabetic retinopathy. Grade 0 is healthy and grade 1 to 4 are diabetic retinopathy with increasing severity, giving 1805 normal images and 1,857 anomalous images. For data balance in the test set, we randomly selected 1,000 normal images as our training set and the rest 2,662 images (805 normal, 1,857 anomalous) as the test set. Images are preprocessed to remove the redundant background and then resized to 256 × 256 pixels.

Br35H is a brain MRI dataset on Kaggle [23]. It contains 1,500 2-D image slices that are tumorous and 1,500 images that are non-tumorous. We randomly selected 1,000 non-tumorous images as our training set and the rest 2,000 images as the test set. It is noted that the 2-D images are slices from the original 3-D imaging modality. Images are resized to 256 × 256 pixels.

ISIC is a skin disease dataset, available as task 3 of ISIC2018 challenge [24]. It contains seven classes and we take NV (nevus) class as normal samples and the rest classes as anomalous samples, following [37]. The official training set contains 6,705 normal images. The official validation set is used as test set, which includes 193 images. Images are resized to 256 × 256 and center cropped to 224 × 224 pixels.

Dataset Exclusion: For the dataset used in previous UAD studies, we found that Chest X-ray [21] used in [5], [37], and [40] and HyperKvasir [48] used in [6] contain anomalous images that differ from normal images not only in diseased anomalies but also in the aspect of medically irrelevant artifacts, e.g. calibrations, labels, and different imaging devices, as shown in Fig. 5. It might be caused by difference sources when collecting healthy and unhealthy samples. In our preliminary exploration, we found that UAD methods do pay strong attention to those regions. Though we achieved promising AUC of 89.6% and 98.5% respectively on these datasets, we do not recommend utilizing such datasets for fair comparison.

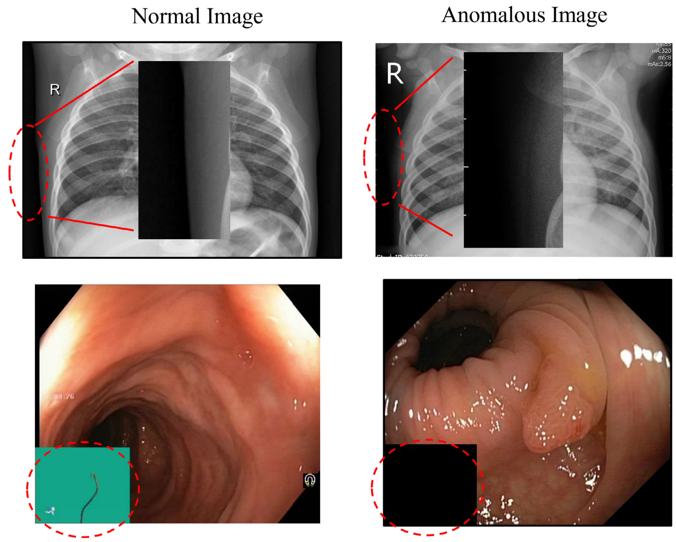


Fig. 5. **Dataset Exclusion**. There are medical-irrelevant differences between normal and anomalous images in some datasets. There are calibration marks in most pneumonia X-rays, but not in healthy X-rays in Chest X-ray (first row). The left-down corner of normal and poly-p endoscope images is in different colors in HyperKvasir (second row).

B. Expermiental Settings and Metrics

Standard ResNet50 pre-trained on ImageNet is utilized as the encoder by default. The decoder is the upside-down version of the encoder without layer 4, i.e., the down-sampling convolution with stride 2 at the beginning of each layer is replaced by an up-sampling interpolation followed by a convolution with stride 1. AdamW optimizer [49] is utilized with β=(0.9,0.999) and weight decay=1e-4. The learning rates (lr) of the encoder and decoder are separately searched in {1e-3, 5e-4, 1e-4} and {1e-4, 5e-5, 1e-5} on each dataset, while keeping the lr of decoder lower than the lr of encoder. The selected encoder *lr* are 5e-4, 5e-4, 5e-4, 1e-4 for OCT2017, APTOS, Br35H, and ISIC, respectively. The selected decoder *lr* are 1e-5, 1e-5, 5e-5, 1e-5, respectively. The networks are trained with the batch size of 32 for 1,000, 4,000, 6,000, and 500 iterations on APTOS, Br35H, OCT2017, and ISIC. In ISIC, normal and anomalous images are both with lesions and they differ by their overall appearance; in the other three datasets, anomalous images differ from normal images by local lesions on the healthy background. Therefore, we take the mean of $\mathcal{M}^{ano}$ as the image anomaly score (indicates overall anomaly) for ISIC and the maximum value of $\mathcal{M}^{ano}$ (indicates local anomaly) for the other three datasets. By default, all batchnorm layers (BN) is in *train* mode during training. However, we observe instability when setting the encoder BN to *train* mode on OCT2017 (probably because of the contrasting white background); therefore, we set it to *eval* on OCT2017. Experiments are conducted on NVIDIA GeForce RTX3090 GPUs and PyTorch.

We adopt Area Under Receiver Operation Characteristic (AUC), F1-score (F1), accuracy (ACC), sensitivity (SEN), and specificity (SPE) as the evaluation metrics. Their definitions are given as followed:

$$\begin{aligned} \text{SEN} &= TP/(TP+FN) \quad SPE \\ &= TN/(FP+TN) \\ \text{PRE (precision)} &= TP/(TP+FP) \\ \text{ACC} &= (TP+TN)/(TP+TN+FP+FN) \\ \text{F1} &= 2 * PRE * SEN/(PRE+SEN) \end{aligned}$$

where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively. AUC is the area under SEN - (1−SPE) curve across different operating thresholds. AUC, F1, and ACC are the evaluation of overall performance, while SEN and SPE focus on the positive samples and negative samples, respectively. Following [5] and [37], the operating threshold of F1, ACC, SEN, and SPE is determined by the optimal value of F1.

C. Comparison to State-of-the-Arts

We compare our method with previous state-of-the-arts (SOTA) on four medical datasets. Auto-Encoder and VAE [9] are basic pixel reconstruction methods. GANomaly [29], f-AnoGAN [7], and SALAD [5] are pixel reconstruction methods based on GAN. ProxyAno [41] is based on auto-encoder with memory bank. SCVAE [43] is based on VAE. MBU-Net [50] is specifically designed for OCT images via layer segmentation. SSD [39] pre-train a domain-specific feature encoder on normal samples via contrastive learning. The aforementioned methods train networks from scratch on the target training set, while the following methods utilize pre-trained encoders. STFPM [32] and MKD [31] are feature distillation methods. RD4AD [11] is a feature reconstruction method. PaDiM [15] is a memory matching method. DifferNet [35], Fastflow [36], and AE-flow [37] are methods based on normalizing flow. We take the results of compared methods from the original papers or papers that report the performances if available. Otherwise, we reproduce the method if code is open source. The results of our EDC are the averages of the last iteration of three runs. Some works report their best performance during training [37]; therefore, we report our best checkpoint (in AUC, check every 100 iterations) during training as well for fair comparison (denoted with †).

TABLE I
ANOMALY DETECTION PERFORMANCE (%) ON OCT2017

Methods	Pub.	AUC	F1	ACC	SEN	SPE
Auto-Encoder	-	77.79	85.79	78.28	94.38	41.70
VAE [9]	ICLR'14	80.04	85.55	77.63	95.32	37.45
GANomaly [29]	ACCV'18	84.02	88.65	81.59	95.86	38.8
f-AnoGAN [7]	MIA'19	83.35	84.73	77.50	89.89	49.36
SALAD [5]	TMI'21	96.42	93.42	90.64	95.69	79.15
ProxyAno [41]	TMI'21	93.30	72.50	84.90	N/A	N/A
SSD [39]	ICLR'21	92.28	91.91	87.20	97.06	57.60
STFPM [32]	BMVC'21	96.86	95.79	93.70	95.59	87.99
MKD [31]	CVPR'21	96.72	94.57	91.60	97.62	73.61
PaDiM [15]	ICPR'21	96.88	95.24	92.80	96.13	82.80
MBU-Net [50]	BSPC'22	N/A	N/A	94.90	97.90	86.00
SCVAE [43]	CIBM'23	94.14	N/A	86.92	N/A	N/A
EDC (Ours)	-	**99.56**	**98.60**	**97.90**	**98.67**	**95.60**
Fastflow [36]†	Arxiv'22	94.08	92.60	88.43	95.45	67.36
AE-flow [37]†	ICLR'23	98.15	96.36	94.42	95.45	88.02
EDC (Ours) †	-	**99.70**	**98.86**	**98.30**	**98.53**	**97.60**

†Results reported with best performance during training. N/A: not available.

1) Results on OCT2017: We first evaluate our method on OCT2017. The experimental results are presented in Table I. Among the compared methods, GAN-based methods outperform AE-based methods among pixel reconstruction paradigms. In addition, methods that make use of pre-trained feature encoders (STFPM, MKD, PaDiM, AE-flow and Ours) generally outperform methods that train networks from scratch (AE, VAE, GANomaly, f-AnoGAN, SALAD, ProxyAno, SCVAE, MBU-Net, SSD), indicating that medical UAD benefits from large-scale pre-training. The advanced methods proposed for general and industrial anomaly detection (STFPM, MKD, and PaDiM) also generalize promisingly well on OCT2017.

Our method producesan superior AUC of 99.56%, F1 of 98.6%, and ACC of 97.9%, exceeding previous SOTA AE-flow by 1.41%, 2.24%, and 3.48%. In the aspect of error (1)-AUC), we achieve 0.44%, reducing the previous SOTA error by **76.2**%.

2) Results on APTOS: Anomaly detection on APTOS is a significantly harder task, as the lesions in retinal fundus images can be extremely inconspicuous, especially for mild and moderate diabetic retinopathy. The experimental results are presented in Table II. Among competing methods, PaDiM cannot perform as well as in OCT2017 because it models the features based on the position. However, fundus images are not registered with the same anatomical position. Though PaDiM produces the best SEN, it sacrifices the SPE with a low anomaly threshold. Surprisingly, other methods for general anomaly detection outperform all previous medical anomaly detection methods, suggesting that researchers should also pay close attention to the latest UAD approaches for natural images while developing medical UAD methods.

Nevertheless, our EDC achieves the best results in all metrics except SEN. Specifically, EDC outperforms the runner-up counterparts STFPM by 1.82% in AUC, 1.2% in F1, and

TABLE II
ANOMALY DETECTION PERFORMANCE (%) ON APTOS

Methods	Pub.	AUC	F1	ACC	SEN	SPE
Auto-Encoder	-	76.91	83.61	74.07	94.83	26.21
VAE [9]	ICLR'14	77.21	85.04	7.12	93.26	39.87
GANomaly [29]	ACCV'18	83.72	85.85	79.37	89.71	55.52
f-AnoGAN [7]	MIA'19	89.64	89.71	85.23	92.29	68.94
SALAD [5]	TMI'21	78.46	85.06	77.35	78.76	42.48
STFPM [32]	BMVC'21	93.59	91.86	88.35	94.23	74.78
MKD [31]	CVPR'21	88.51	81.70	87.84	94.72	51.67
PaDiM [15]	ICPR'21	77.17	85.47	77.24	**95.96**	34.04
RD4AD [11]	CVPR'22	92.43	90.65	86.44	94.18	68.57
EDC (Ours)	-	**95.41**	**93.06**	**90.08**	93.81	**81.12**
EDC (Ours) †	-	95.71	92.88	89.97	95.37	77.89

†Results reported with best performance during training

TABLE III
ANOMALY DETECTION PERFORMANCE (%) ON BR35H

Methods	Pub.	AUC	F1	ACC	SEN	SPE
Auto-Encoder	-	89.57	89.98	84.20	94.66	52.80
VAE [9]	ICLR'14	91.36	91.24	86.80	91.73	72.00
GANomaly [29]	ACCV'18	93.78	92.85	89.10	94.40	73.20
f-AnoGAN [7]	MIA'19	94.80	93.76	90.45	95.66	74.80
SALAD [5]	TMI'21	97.69	96.83	95.15	98.86	96.10
STFPM [32]	BMVC'21	93.11	93.46	89.8	97.20	67.60
MKD [31]	CVPR'21	98.70	94.30	96.26	98.06	83.00
PaDiM [15]	ICPR'21	98.19	98.00	96.95	99.80	88.40
RD4AD [11]	CVPR'22	99.12	99.04	98.55	99.93	94.40
EDC (Ours)	-	**99.85**	**99.57**	**99.35**	**99.95**	**97.40**
EDC (Ours) †	-	99.95	99.47	99.20	100	96.80

†Results reported with best performance during training

1.73% in ACC. In the aspect of AUC error, we achieve 4.59%, reducing the previous best error by **28.4**%. Though EDC does not perform best in SEN, it produces a favorable SEN-SPE trade-off.

3) Results on Br35H: Anomaly detection on Br35H is a relatively simple task, as tumors are generally conspicuous in this MRI dataset. The experimental results are presented in Table III. The performance of SOTA methods are comparatively good, with four methods (MKD, PaDiM, RD4AD and Ours) exceeding 98% in AUC.

Our method produces an excellent AUC of 99.85%, F1 of 99.57%, and ACC of 99.35%, outperforming previous SOTA RD4AD by 0.73%, 0.53%, and 0.8%. In the aspect of AUC error, our EDC presents only 0.15%, reducing the previous error by **83**%.

4) Results on ISIC: For the ISIC skin lesion image dataset, the normal class (nevus) is a kind of benign lesion, instead of a completely normal object like other datasets (plain skin in this scenario), which makes normal images easily mistaken with anomalous images. The experimental results are presented in Table IV. All compared methods present relatively undesirable results (AUC under 90%), including the most recently proposed AE-flow in which ISIC is first proposed as a UAD dataset. Methods based on normalizing flow generally outperform GAN-based pixel reconstruction methods, which may also be attributed to the pre-trained feature extractor utilized in such methods.

The results of our last iteration are already comparable to previous SOTA AE-flow that reports their best results during training. In the aspect of best checkpoints, we achieve superior results in all metrics compared to AE-flow. However,

TABLE IV
ANOMALY DETECTION PERFORMANCE (%) ON ISIC

Methods	Pub.	AUC	F1	ACC	SEN	SPE
PaDiM [15]	ICPR'21	82.13	72.04	73.06	**95.71**	60.16
SSD [39]	ICLR'21	79.67	71.25	76.17	81.43	73.17
RD4AD [11]	CVPR'22	76.76	70.47	67.43	84.29	62.60
EDC (Ours)	-	**89.33**	**80.10**	**84.28**	84.76	**84.01**
Auto-Encoder †	-	73.93	62.57	64.77	78.57	56.91
GANomaly [29]†	ACCV'18	72.54	60.95	56.99	90.00	38.21
f-AnoGAN [7]†	MIA'19	79.80	67.03	68.39	85.71	58.54
DifferNet [35]†	WACV'21	76.52	64.85	64.25	87.14	51.22
Fastflow [36]†	Arxiv'21	80.81	69.96	90.98	90.00	86.99
AE-flow [37]†	ICLR'23	87.79	80.56	84.97	81.43	86.99
EDC (Ours)†	-	**90.53**	**81.94**	**86.53**	**84.29**	**87.80**

†Results reported with best performance during training

the performance is still less satisfactory than other image modalities. We argue that the setting of UAD is designed for detecting anomalies in healthy objects, which is not suitable for the classification of different diseases or lesions. Most UAD methods construct the anomaly detection paradigm by segmenting unseen regions (lesions) on seen backgrounds (objects); however, it is less reasonable to segment unseen lesions out of normal lesions. Besides, the intra-class variation is more significant than other image modalities, causing the false-positive activation of irrelevant objects like hairs and pores. Therefore, such tasks require investigation to further improve performance.

D. Visualization of Anomaly Localization

Localization of lesions is important to the explanation of detection in clinical disease screening. We present the anomaly maps of anomalous samples by min-max-normalizing each $\mathcal{M}^{ano}$ to 0-1 and overlaying it on the input image. The visualization result is shown in Fig. 6. Our method produces strong responses to diseased regions. It is also observed that EDC detects large lesions by their edge. However, the anomaly map of the anomalous image in ISIC is less instructive, activating lesions as well as other noises, which accounts for the unfavorable anomaly detection performance.

E. Ablation Study

In this section, we conduct further ablation studies on APTOS and OCT2017 to evaluate the contribution made by each proposed element, as shown in Table V. Each result is reported with a single run. Because some configurations are extremely unstable in training, we report both the last and the best AUC during training. Starting from conventional feature reconstruction (*config.* 1), optimizing the encoder and decoder together (*config.* 2) causes a small performance drop on APTOS but a complete collapse on OCT2017. By introducing the stop-gradient operation to construct a contrastive learning paradigm (*config.* 3), the performance is recovered and further improved. Finally, replacing point-wise cosine distance with the proposed global cosine distance yields the best result (*config.* 4, Ours). In addition, we show that the global cosine distance can improve the baseline feature reconstruction alone (*config.* $1 + \mathcal{L}_{global}$), indicating the generality of the proposed objective function.

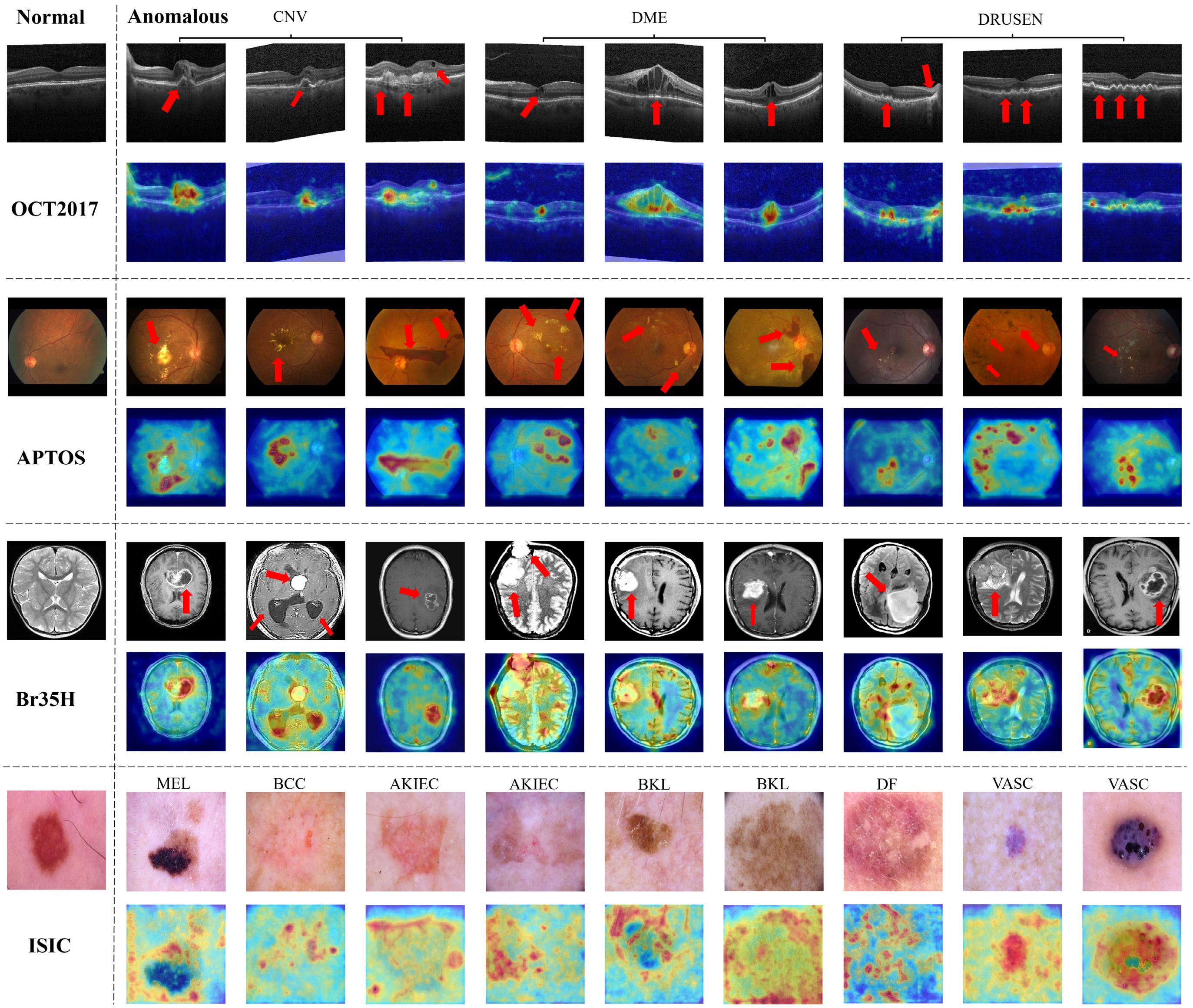


Fig. 6. Visualization of anomaly maps. The first column from the left presents normal images as references. On OCT2017, APTOS, and Br35H, anomalous regions are indicated by red arrows. On ISIC, the whole foreground lesions are considered to be anomalous.

TABLE V
ABLATION STUDY ON APTOS AND OCT2017

Configuration	*optimize encoder*	*stop-gradient*	$\mathcal{L}_{global}$	APTOS AUC(%)	APTOS AUC†(%)	OCT2017 AUC(%)	OCT2017 AUC†(%)
config. 1	-	-	-	91.18	91.18	92.59	98.98
config. 2	✓	-	-	87.00	89.28	43.25	57.35
config. 3	✓	✓	-	94.24	94.26	98.39	99.47
config. 1 + $\mathcal{L}_{global}$	-	-	✓	94.58	94.58	99.25	99.37
config. 4 (Ours)	✓	✓	✓	**95.62**	**95.62**	**99.69**	**99.79**

†Results reported with best performance during training

By default, the predicted anomaly map $\mathcal{M}^{ano}$ is the ensemble of all anomaly maps $\mathcal{M}^1$, $\mathcal{M}^2$, and $\mathcal{M}^3$. We further explore the detection performance of the anomaly map of each layer, as shown in Table VI. We observe that different dataset favors different layer. $\mathcal{M}^3$ yields the best result in APTOS, while $\mathcal{M}^2$ yields the best result in OCT2017. As each layer has a different receptive field, they are sensitive to different kinds and sizes of anomalies. The superior result is obtained by ensemble the anomaly maps of all layers to detect a wider range of anomalies.

In addition, we test a variety of encoder backbones, i.e., ResNet50 (default), ResNet34, WideResNet50 [51], and ResNeXt50 [52]. The corresponding decoder is the reversed version of the encoder. It is noted that different encoders

TABLE VI
PERFORMANCE (AUC, %) OF ANOMALY MAPS OF DIFFERENT LAYERS

$\mathcal{M}^k$ for $\mathcal{M}^{ano}$	APTOS	OCT2017
$\mathcal{M}^1$	92.85	98.35
$\mathcal{M}^2$	88.85	99.59
$\mathcal{M}^3$	95.32	98.51
$\mathcal{M}^1$, $\mathcal{M}^2$, $\mathcal{M}^3$	**95.62**	**99.69**

TABLE VII
PERFORMANCE (AUC, %) OF DIFFERENT ENCODER BACKBONES

encoder backbone	APTOS	OCT2017
ResNet50 (default)	95.62	99.69
ResNet34	94.96	98.98
ResNeXt50-32x4d	94.50	**99.77**
WideResNet50	**96.47**	99.50

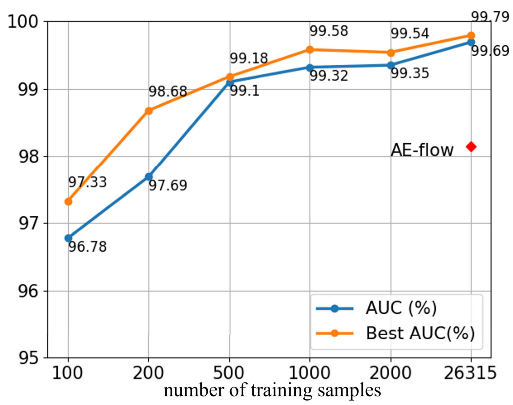


Fig. 7. Performance on OCT2017 with varied numbers of training images.

may favor different optimal learning rates. We set the encoder learning rate of ResNet50, ResNet34, ResNext50 to 1e-5 (default), and WideResNet50 to 5e-5. Results are shown in Table VII. Different backbones produce comparable results, suggesting the generality of our method.

F. Training With Fewer Samples

Building UAD models with a limited number of normal samples is a relevant setting for real-world scenarios, e.g., after a piece of new imaging equipment is launched in a hospital. Therefore, we study the performance with fewer training images. On OCT2017 with 26315 normal samples, we vary the number of training samples from 100 (0.38%) to 2000 (7.6%). As training with fewer samples for the same iterations (6000) causes over-fitting, we also present the best AUC during training to save searching time. Results are presented in Fig. 7. With only 200 (0.76%) normal samples, our EDC can still match previous SOTA performance.

V. CONCLUSION

In this study, we propose a novel contrastive learning paradigm, namely EDC, for medical anomaly detection. It enhances the transfer ability of the pre-trained encoder by jointly optimizing all parameters end-to-end without pattern collapse and training instability. In addition, we propose a new objective function, namely global cosine distance, to make the contrast between feature point manifolds to stabilize the optimization and further improve performances. Extensive experiments demonstrate our superiority, setting a new state-of-the-art on four public datasets. The proposed EDC with 2-D encoder is currently limited to 2-D input. For application on 3-D modalities, our method can be applied to the 2-D slices of 3-D images and the results are aggregated. The proposed method also has the potential to extend to 3-D input given a pre-trained 3-D encoder. The idea of optimizing encoders may further boost the research of UAD methods not only in medical images but also in other image applications that are far from natural image domain.

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